

IT 520 · Oct 7 activity: what broke at receiving?

Name: _____

Tote 9's packing record ships to Zürich. Work with a partner, and use the reference box.

UTF-8 byte count (by code point)

Code point	Bytes
U+0000 to U+007F	1
U+0080 to U+07FF	2
U+0800 to U+FFFF	3

Reading bytes back: a byte 00–7F is one whole character; C2–DF starts a 2-byte character; E0–EF starts a 3-byte character; 80–BF continues the character before it.

File signatures (first bytes, hex)

Format	Starts with
PNG image	89 50 4E 47 0D 0A 1A 0A
PDF document	25 50 44 46 2D
ZIP archive	50 4B 03 04

Three agreements: the **bytes** (unchanged?), the **encoding** (bytes to characters), the **format** (how the file is organized).

Part A: Text and bytes

1. **Predict.** The packing app saves Zürich in UTF-8. Code points: Z = U+005A, ü = U+00FC, and r, i, c, h are ASCII. How many characters? How many bytes? Show the sum.
2. **Reverse.** A file holds these UTF-8 bytes: 4E 69 C3 B1 6F. How many characters do they spell? Which bytes belong to the same character, and how do you know?
3. **Fields.** The header is tote,item,dest. A reader gets the line 9,Scarf, red,Zürich. How many fields does it find? Rewrite the line so it has three.

Part B: Diagnose. Name the broken agreement and give the evidence.

4. **File 1.** Receiving opens tote9.csv and sees ZÃ¼rich. The destination's bytes are 5A C3 BC 72 69 63 68, and the file's SHA-256 digest matches the sender's.
5. **File 2.** Receiving's CSV reader rejects manifest.csv. Its first eight bytes are 25 50 44 46 2D 31 2E 37. What is this file, and what should receiving do?
6. **Two copies.** Receiving has two copies of the export and the sender's published digest.

File	SHA-256 digest
Sender's tote9.csv	c7d6ca3634dcf45a313c24d7e2e8ebd7ac6538d48557fc1d61f1e24bb5ab4b4a
Copy A: received.csv	c7d6ca3634dcf45a313c24d7e2e8ebd7ac6538d48557fc1d61f1e24bb5ab4b4a
Copy B: tote9.csv	85aaa6eee3b00f53ecca03f4d7cf3ed4122d1cfdc78d30592b70ad5530e10da9

(a) Which copy's bytes changed? (b) Copy A has a different name. Why does its digest still match? (c) Copy A matches. Could receiving still display it wrong? Explain using File 1.